



MATH 241

Fall 2017

Quiz 4

27/11/17

DURATION: 60 MINUTES

Name : _____

Student ID : _____

Signature : _____

SOLUTION KEY

This quiz contains 4 pages (including this cover page) and 3 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You are required to show your work on each problem on this quiz. The following rules apply:

- This is a single exit exam, once you leave the exam place, you can not come back.
- You must not leave without signing the exam attendance sheet.
- You can not leave in the first 20 minutes.
- Show your student ID to your proctor while signing the attendance sheet.
- Calculators, mobile phones and smart watches are not allowed
- Mysterious or unsupported answers will not receive credits, explain every step in your solutions.
- If you need more space, use the back of the pages; clearly indicate when you have done this.

Problem	Points	Score
1	35	
2	35	
3	30	
Total:	100	

Do not write in the table to the right.

1. (35 points) Solve the difference equation

$$x[n] = x[n-1] - \frac{1}{4}x[n-2]$$

with the initial conditions $x[-1] = 1$ and $x[-2] = -1$

Let $x[n] = A \cdot w^n$, where A and w are constants.

$$x[n-1] = A \cdot w^{n-1}$$

$$x[n-2] = A \cdot w^{n-2}$$

$$\Rightarrow A \cdot w^n - A \cdot w^{n-1} + \frac{1}{4} A \cdot w^{n-2} = 0$$

$$1 - w^{-1} + \frac{1}{4} w^{-2} = 0 \Rightarrow w = \frac{1}{2} \text{ double root.}$$

Now, $x[n] = (A_1 + A_2 \cdot n) \cdot w^n$

$$= (A_1 + A_2 \cdot n) \left(\frac{1}{2}\right)^n$$

with A_1, A_2 as constants to be found with $x[-1] = 1$
 $x[-2] = -1$

$$\Rightarrow x[-1] = (A_1 - A_2) \left(\frac{1}{2}\right)^{-1} = (A_1 - A_2) 2 = 1$$

$$A_1 - A_2 = \frac{1}{2}$$

$$x[-2] = (A_1 - 2A_2) \cdot \left(\frac{1}{2}\right)^{-2} = (A_1 - 2A_2) \cdot 4 = -1$$

$$A_1 - 2A_2 = -\frac{1}{4}$$

$$A_1 = \frac{5}{4}$$

$$A_2 = \frac{3}{4}$$

$$\Rightarrow x[n] = \left(\frac{5}{4} + \frac{3}{4}n\right) \cdot \left(\frac{1}{2}\right)^n$$

2. (35 points) Find canonical form of

$$f(x_1, x_2) = \frac{x_1^2}{2} - 3x_1x_2 + \frac{x_2^2}{2}$$

and find the values of (x_1, x_2) such that $f(x_1, x_2) \geq 0$.

$$f(x_1, x_2) = [x_1 \ x_2] A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1/2 & -3/2 \\ -3/2 & 1/2 \end{bmatrix}$$

$$\lambda_1 = 2$$

$$\bar{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$\lambda_2 = -1$$

$$\bar{v}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

normalized eigenvectors

$$\Rightarrow Q = [\bar{v}_1 \ \bar{v}_2] = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$Q^T = Q^{-1}$$

$$\Rightarrow \bar{A} = Q \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} Q^T$$

$$\text{and } f(x_1, x_2) = [x_1 \ x_2] \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\boxed{f(x_1, x_2) = (x_1 - x_2)^2 - \frac{1}{2}(x_1 + x_2)^2} \quad \text{canonical form}$$

$$\text{for } \underline{f(x_1, x_2) \geq 0}, \quad (x_1 - x_2)^2 \geq \frac{1}{2}(x_1 + x_2)^2$$

$$|x_1 - x_2| \geq \frac{1}{\sqrt{2}} |x_1 + x_2|$$

$$\underline{\sqrt{2} |x_1 - x_2| \geq |x_1 + x_2|}$$

So, looking for the different cases for $|x_1 - x_2|$ and $|x_1 + x_2|$ we get;

2. (35 points) Find canonical form of

$$f(x_1, x_2) = \frac{x_1^2}{2} - 3x_1x_2 + \frac{x_2^2}{2}$$

and find the values of (x_1, x_2) such that $f(x_1, x_2) \geq 0$.

continues...

$$f(x_1, x_2) \geq 0 \text{ when } \left\{ \begin{array}{ll} x_1 > \frac{\sqrt{2}+1}{\sqrt{2}-1} x_2, & x_1 > x_2 > 0 \\ & x_1 > 0 > x_2 \text{ and } |x_1| > |x_2| \\ x_1 > \frac{\sqrt{2}-1}{\sqrt{2}+1} x_2, & x_1 > 0 > x_2 \text{ and } |x_1| < |x_2| \\ & 0 > x_1 > x_2 \\ x_1 \leq \frac{\sqrt{2}-1}{\sqrt{2}+1} x_2, & x_2 > x_1 > 0 \\ & x_2 > 0 > x_1 \text{ and } |x_1| < |x_2| \\ x_1 \leq \frac{\sqrt{2}+1}{\sqrt{2}-1} x_2, & x_2 > 0 > x_1 \text{ and } |x_1| > |x_2| \\ & 0 > x_2 > x_1 \end{array} \right.$$

3. (30 points) Prove

1. If $\lambda \neq 0$ is an eigenvalue of A , then $\frac{1}{\lambda}|A|$ is an eigenvalue of $\text{adj}(A)$.
2. If v is an eigenvector of A , then v is also an eigenvector of $\text{adj}(A)$.

$$\leftarrow |A| = \det(A)$$

Using the fact that $A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \det(A) \cdot I$

$$Ax = \lambda x \quad \text{--- Eq(1)}$$

$$\underbrace{\text{adj}(A)}_{\det(A)} \cdot Ax = \text{adj}(A) \cdot \lambda \cdot x$$

multiplying $\text{adj}(A)$ from the left

$$\det(A) \cdot x = \lambda \cdot \text{adj}(A) \cdot x$$

λ is a scalar and can be moved to the left

$$\Rightarrow \frac{\det(A)}{\lambda} \cdot x = \text{adj}(A) \cdot x \iff \text{adj}(A) \cdot x = \underbrace{\frac{\det(A)}{\lambda}}_{\text{eigenvalue}} \cdot x \quad \text{--- Eq(2)}$$

1) Since $\text{adj}(A) \cdot x = \frac{\det(A)}{\lambda} \cdot x$, λ is a scalar constant and $x \neq 0$ (where x is an eigenvector of A corresponding to λ), $\frac{\det(A)}{\lambda}$ is an eigenvalue of $\text{adj}(A)$

2) Since Eq(2) is derived from Eq(1), both has the same eigenvector.